Husnu S. Narman

: http://hsnarman.github.io
: http://www.linkedin.com/in/husnunarman
: https://bit.ly/2KQCc1x

narman@marshall.edu
Office : (304) 696-5829

ACADEMIC EXPERIENCE

Associate Professor: Marshall University — **Aug 2023 - Present**
Department of Computer Sciences and Electrical Engineering — *Huntington WV, USA*

Assistant Professor: Marshall University — **Aug 2017 - May 2023**
Department of Computer Sciences and Electrical Engineering — *Huntington WV, USA*

Instructor: Marshall University — **Jan 2017 - May 2017**
Department of Computer Sciences and Electrical Engineering — *Huntington WV, USA*

Postdoctoral Fellow: Clemson University — **May 2016 - Dec 2016**
Department of Electrical and Computer Engineering — *Clemson SC, USA*

EDUCATION

Ph.D. in Computer Science — **May 2016**
University of Oklahoma — *Norman OK, USA*

M.S. in Computer Science — **May 2011**
University of Texas at San Antonio — *San Antonio TX, USA*

B.S. in Mathematics — **Aug 2006**
Abant Izzet Baysal University — *Bolu, Turkey*

HONORS

Weisberg Service Award
College of Engineering and Computer Sciences, Marshall University — *2021 - 2022*

Distinguished Artists and Scholars Junior Category Award
Marshall University — *2020 - 2021*

Weisberg Academy of Distinguished Teachers Award
College of Engineering and Computer Sciences, Marshall University — *2020 - 2021*

Outstanding Service Award
IEEE/ACM IoTDI — *2021*

Nomination for Pickens Queen Teaching Award
Marshall University — *2018, 2021*

Outstanding Ph.D. Student in Computer Science
University of Oklahoma — *2015 - 2016*

Study Abroad Fellowship for Higher Education
Turkish Ministry of National Education — *2007 - 2016*

Graduation as High Honor Student
Abant Izzet Baysal University — *2006*

RESEARCH INTEREST

Advanced Learning Technologies: *Investigate usage of high-tech products such as Virtual and Augmented Reality based applications with their effects in learning for K-12 and higher education.*

Data Mining: *Using sentiment analysis tools to understand the user behaviors for various applications like shopping habits and chain effects.*

Resource Allocation: *Determine scheduling models and allocations policies in Cloud and Fog Computing, IoT, and Networks with their applicable cases such as Vehicular Networks and Crowdsourcing.*

Smart Health: *Develop and investigate IoT products and mobile applications with machine learning algorithms to make the health system more accessible and productive by helping doctors and patients.*

STEM Education: *Develop applications for K-12 and higher education to overcome difficulties in learning Math and Coding subjects.*

R&D GROUP ATTAINMENTS

Fellowship & Scholarship

- Josh Maddy (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2023; *Project: The Metaphysical Exhibition An Exploration of Technology, the Arts, and Sciences: Expansion*
- Neil Loftus (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2023; *Project: Detecting Birds with Real Time Image Processing for Drone Safety: Expansion*
- Neil Loftus (Undergraduate): NASA Undergraduate Research Fellowship 2022 - 2023; *Project: Detecting Birds with Real Time Image Processing for Drone Safety*
- Neil Loftus (Undergraduate): Marshall University Creative Discovery and Research Award for Summer 2022; *Project: The Cybersecurity Packet Control Simulator: The Effect of Visual Learning Tools on Retention of Information in Computer Science*
- Josh Maddy (Undergraduate): Marshall University Creative Discovery and Research Award for Summer 2022; *Project: The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences*
- Josh Maddy (Undergraduate): Marshall University SURE Summer Research Fellowship 2022; *Project: Augmented Reality as an Aid for Physics Concepts*
- Neil Loftus (Undergraduate): Marshall University SURE Summer Research Fellowship 2022; *Project: The Cybersecurity Packet Control Simulator*
- Eric Dillion (Undergraduate): Marshall University Creative Discovery and Research Award for Spring 2022; *Project: Automatic Feedback System to Teach Cybersecurity Principles*
- Eric Shoemaker (Undergraduate): Marshall University Creative Discovery and Research Award for Summer 2021; *Project: Crowdsourcing based Community Infrastructure Management Application*
- Eric Shoemaker (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2021; *Project: Community Infrastructure Management Application*
- Jarred Carter (Undergraduate): NASA Undergraduate Research Fellowship 2020 - 2021; *Project: Simulation for Trade-off Model of Fog-Cloud Computing for Space Network*
- William Coleman (Undergraduate): Marshall University SURE Summer Research Fellowship 2020; *Project: Enhancing STEM Education with Augmented Reality*
- James Farley (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2020; *Project: Emotion Classification of Users in Social Media*
- Alex Canfield, Cameron Berry, Jeremy Giese, Logan Carpenter (Undergraduates): Marshall University Research Scholar Award for Spring 2019; *Project: Data Structure with Augmented Reality*
- James Farley (Undergraduate): Marshall University Creative Discovery and Research Award for Fall 2019; *Project: Emotion Classification of Users based on the Comments and Emojis in Social Media*
- Jarred Carter (Undergraduate): NASA Undergraduate Research Fellowship 2019 - 2020; *Project: Trade-off Model of Fog-Cloud Computing for Space Network*
- Caleb Kesler (Undergraduate): Marshall University Creative Discovery and Research Award for Summer 2019; *Project: Development of Story-Assisted Platform for Early Childhood for Coding*
- Alymbek Damir Uulu (Undergraduate): Marshall University SURE Summer Research Fellowship 2019; *Project: Profile Analysis on Cryptocurrency Investors and Social Engineering on Their Prices*

- Jared Lee Lewis (Undergraduate): Marshall University Creative Discovery and Research Award for Fall 2018; *Project: Automated IP Reputation Analyzer System with Machine Learning*
- Geanina Tambaliuc (Undergraduate): Marshall University SURE Summer Research Fellowship 2018; *Project: Automated IP Reputation Analyzer System*
- Alex Kacinari, Chris Murphy, Derek M Staley (Undergraduates): Marshall University Research Scholar Award for Spring 2018; *Project: Suturing Technique Simulation*
- Charlie Murphy, Michael B Branard, Steven D. Gunnels (Undergraduates): Marshall University Research Scholar Award for Spring 2018; *Project: Embedded Storybook Game*

Internship/Development

- Hwapyeong Song (Graduate): WV Department of Education Internship and Paving Academy 2020
- Neil Loftus (High School): Paving Academy 2020
- Geanina Tambaliuc (Undergraduate): WV Department of Education Internship 2019
- Anh Nguyen (Undergraduate): WV Department of Education Internship 2019
- Jake Gressang (Undergraduate): WV Department of Education Internship 2019
- Kuo Chi Fang (Graduate): WV Department of Education Internship 2018
- Ibrahim Hussein Mwinyi (Graduate): WV Department of Education Internship 2018

Publications

- High School Students: Isabella Schrader, Laina Karim, Neil Loftus, Sawyer Slack.
- Undergraduate Students: Alex Canfield, Alymbek Damir Uulu, Anh Nguyen, Cameron Berry, Cameron Green, Craig Carpenter II, Eric M. Dillon, Eric Shoemaker, Geanina Tambaliuc, Greg Weed, James Farley, Jared Lee Lewis, Jarred Carter, Jeremy Giese, John Cook, Josh Maddy, Logan Carpenter, Neil Loftus, Thomas D. Wills, William Coleman.
- Graduate Students: Govind Yatnalkar, Ibrahim Hussein Mwinyi, Kanimozhi Kalaichelavan, Kuo Chi Fang, Sreehari Sreenath.

TAUGHT COURSES AND TEACHING EFFECTIVENESS

Q1	I believe that I learned in this class.
Q2	The course was well organized.
Q3	This course challenged me intellectually.
Q4	I have become more competent in this area because of this course.
Q5	The objectives of the course were well explained.
Q6	The instructor followed his/her syllabus.
Q7	The instructor gave clear explanations to clarify concepts.
Q8	The instructor was supportive in academic situations.
Q9	The instructor showed enthusiasm when teaching.
Q10	The instructor informed students of their progress.
Q11	The instructor's use of examples helped to get points across in class.
Q12	The instructor adequately explained the grading scale.
Q13	The instructor treated me fairly.
Q14	The instructor was enthusiastic about the course material.
Q15	The instructor encouraged students to ask questions.
Q16	The instructor provided me with an effective array of challenges.
Q17	The instructor carefully answered questions raised by students.
Q18	The instructor treated students with respect.
Q19	The instructor presented material in a clear manner.
Q20	The instructor used class time well.
Q21	The instructor seemed genuinely interested in wanting me to learn.
Q22	I would recommend this instructor to other students.

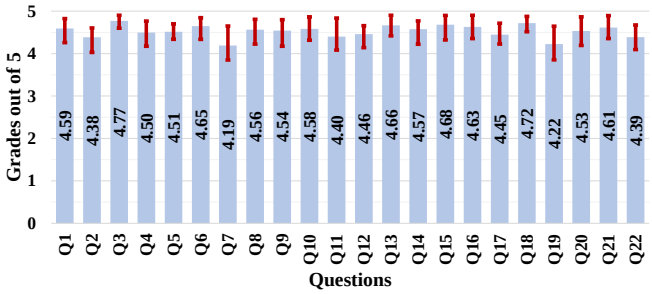

Figure 1: Each question (left table) is asked to more than 429 undergraduate and graduate students to grade from 1 to 5 in course evaluations. The figure bars show the average grades with minimum and maximum values for each question (the average is out of 5).

Table 1: *Enrollment* shows the number of registered students, *R.Rate* shows the percentage of students who answered the course evaluation; and scores show the mean score out of 4 for each course and semester.

Term	Course	E.ment	R.Rate	C.Score	S.Score
Spring 2017	CS 210 Data Structures and Algorithms	12	75%	3.64	3.49
	CS 215 Adv. Data Structures and Algorithms	17	65%	3.22	
	CS 410 Database Engineering	23	83%	3.28	
	CS 510 Database Systems	29	76%	3.75	
	IS 623 Database Management	13	85%	3.72	3.72
Fall 2017	CS 210 Data Structures and Algorithms	20	90%	3.73	3.72
	CS 651 Cloud Computing	16	94%	3.70	
Spring 2018	CS 210 Data Structures and Algorithms	13	85%	3.72	3.77
	CS 510 Database Systems	10	80%	3.83	
	IS 623 Database Management	27	96%	3.77	3.77
Fall 2018	CS 210 Data Structures and Algorithms	16	88%	3.79	3.84
	CS 360 Automata and Formal Languages	21	76%	3.76	
	CS 651 Cloud Computing	18	100%	3.95	
Spring 2019	CS 120 Computer Science II (OOP-Java)	20	75%	3.64	3.63
	CS 210 Data Structures and Algorithms	14	86%	3.60	
	CS 681 Master Thesis	1	100%	3.71	
Fall 2019	CS 120 Computer Science II (OOP-Java)	18	94%	3.65	3.71
	CS 210 Data Structures and Algorithms	12	83%	3.79	
	CS 681 Master Thesis	1	100%	3.89	
Spring 2020	CS 120 Computer Science II (OOP-Java)	39	74%	3.70	3.71
	CS 210 Data Structures and Algorithms	12	92%	3.75	
	CYBR 620 Cyberwarfare	1	100%	3.89	3.89
Fall 2020	CS 210 Data Structures and Algorithms (Vir)	15	100%	3.53	3.67
	CS 210 Data Structures and Algorithms (Web)	6	67%	3.62	
	CS 602 Cloud Computing (Web)	13	92%	3.87	
Spring 2021	CS 210 Data Structures and Algorithms (Vir)	27	81%	3.44	3.54
	CS 510 Advanced Database Systems (Vir)	3	100%	3.98	
	CS 510 Advanced Database Systems (Web)	4	100%	3.75	
	CYBR 620 Cyberwarfare (Web)	7	100%	3.51	3.51
Fall 2021	CS 210 Data Structures and Algorithms	15	93%	3.68	3.66
	CS 210 Data Structures and Algorithms (Web)	14	43%	3.51	
	CS 602 Cloud Computing (Web)	5	80%	3.80	
Spring 2022	CS 210 Data Structures and Algorithms	19	89%	3.59	3.51
	CS 210 Data Structures and Algorithms (Web)	11	55%	3.20	
	CS 510 Advanced Database Systems (Web)	9	78%	3.57	
	CYBR 620 Cyberwarfare (Web)	5	80%	3.71	3.71
Fall 2022	CS 210 Data Structures and Algorithms	19	89%	3.71	3.59
	CS 210 Data Structures and Algorithms (Web)	15	40%	3.34	
	CS 602 Cloud Computing (Web)	7	86%	3.41	
Spring 2023	CS 210 Data Structures and Algorithms	15	73%	3.66	3.51
	CS 210 Data Structures and Algorithms (Web)	15	73%	3.32	
	CS 510 Advanced Database Systems (Web)	11	100%	3.50	
	CS 685 Independent Study	1	100%	4.00	

Table 1 shows the taught courses with student evaluation results for each course and semester. Undergraduate courses are from 1xx to 4xx, and graduate courses are from 5xx and above. Mostly, a high number of students responded the student evaluation. Figure 1 also shows the course evaluations for each question.

STUDENT ADVISING

Table 2 shows the number of advised students in each semester. Graduate students have not been advised between Spring 2018 and Fall 2020 due to policy changes of the Department of Computer Sciences and Electrical Engineering for advising.

Table 2: The number of students advised in each semester.

Term	Total	Undergraduate	Graduate
Spring 2017	18	9	9
Fall 2017	35	27	8
Spring 2018	27	26	1
Fall 2018	41	41	0
Spring 2019	39	39	0
Fall 2019	39	39	0
Spring 2020	27	27	0
Fall 2020	32	30	2
Spring 2021	21	18	3
Fall 2021	36	29	7
Spring 2022	36	28	8
Fall 2022	36	29	7
Spring 2023	33	24	9

SERVICES

Marshall University **Huntington WV, USA**

- Organization of Computer Science Adventure Zone Summer Camp 2017 - Present
- Elected College Representative on the Athletic Committee 2019 - Present
- CS Freshman Orientation 2018 - Present
- CS Online Course Development 2018 - Present
- CS Conference Organization Committee 2018 - Present
- Computer Science (CS) Student Leadership Program Coordinator 2019 - 2020
- CS Representative on the College Outreach and Recruitment Committee 2019 - 2020
- CS Outreach Committee Chair 2017 - 2020
- CS Faculty Candidate Interviewing Committee 2017 - 2019
- Catalog Editing and Correction for BS and MS Programs in CS 2017 - 2018
- CS Project Room Management 2017 - 2018

PROFESSIONAL ACTIVITIES

Co-Lead for Training Programs **2021 - Present**
Institute for Cyber Security (ICS), Marshall University

Editorial Board **2018 - Present**
Editorial Board: Elsevier Journal of Network and Computer Applications

Summer Camp on Robotics and Cybersecurity for K-12 **2017 - Present**
Marshall University *Huntington WV, USA*

Faculty Mentor Internship Program **2018 - 2020**
WV Department of Education *Huntington WV, USA*

Virtual Conference Working Group Lead
• 2021 ACM/IEEE Conference on Internet of Things Design and Implementation

Publicity Co-Chair
• 2017 International Conference on Networking, Architecture, and Storage • 2017 International Conference on Computer Communications and Networks

Technical Program Committee
• IEEE Global Communications Conf. • IEEE Int. Conf. on Communications • Springer Ubiquitous Networking Conf. • IEEE Wireless Communications and Networking Conf. • IEEE Int. Conf. on Communications, Network, and Satellite • IEEE 5G World Forum (WF-5G) • IEEE Int. Conf. on Internet of Things and Intelligence System • Elsevier Int. Conf. on Ambient Systems, Networks and Technologies • IEEE Int. Conf. on Fog and Edge Computing • IEEE Symp. on Signal Processing and Information Technology • IEEE Int, Conf. on Signals and Systems • IEEE/ACM Int. Symp. in Cluster, Cloud, and

Grid Computing • IEEE Middle East & North Africa Communications Conf. • IEEE Int. Conf. on Wireless Networks and Mobile Communications • IEEE TENCON

Journal Reviewer

• IEEE Journal on Selected Areas in Communications • IEEE Transactions on Mobile Computing • IEEE Transactions on Parallel and Distributed Systems • IEEE Transactions on Vehicular Technology • IEEE Transactions on Industrial Electronics • IEEE Transaction on Intelligent Transportation Systems • IEEE Transactions on Sustainable Computing, • ACM Transactions on Cyber-Physical Systems • ACM Transactions on Knowledge Discovery from Data • Elsevier Journal of Network and Computer Applications • Elsevier Future Generation Computer System • MDPI Sensors • Wiley Software: Practice and Experience

Membership

• IEEE Senior Member (2022 - Present) • IEEE Member (2014 - 2021)

VOLUNTEER ACTIVITIES

VEX IQ Robot Tournament for Middle and Elementary Schools **2019 - Present**
Marshall University *Huntington WV, USA*

Faculty adviser for IEEE Marshall University Student Chapter **2022 - Present**
Marshall University *Huntington WV, USA*

Faculty adviser for Geeks and Gadgets Club **2018 - Present**
Marshall University *Huntington WV, USA*

Coordinator and Initiator for Community Service Leadership Program **2019 - 2020**
Marshall University *Huntington WV, USA*[-5pt]

CERTIFICATION

Responsible Conduct of Research **2018**
Collaborative Institutional Training Initiative Program *Huntington WV, USA*

Behavioral & Social Science Research **2018**
Collaborative Institutional Training Initiative Program *Huntington WV, USA*

Independent Improving Your Online Course (IYOC) **2017**
Quality Matters *Huntington WV, USA*

Java SE 7 **2016**
Robert Half Technology *Clemson SC, USA*

GRANTS

External Funded and Under Review Grant Applications

[15] *Automatic Assessment and Repair for Railroads.* **Co-PI**. ERDC, 2023. Amount: $2M, **Funded**.

[14] *Establishing a Cyber Security Center for Critical Infrastructure.* **Responsible for Training Modules**. Department of Education, 2023. Amount: Total:$1,5M, Share: $750K, **Funded**.

[13] *Detecting Birds with Real Time Image Processing for Drone Safety.* **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2022. Amount: $5K, **Funded**.

[12] *Detecting Birds with Real Time Image Processing for Drone Safety: Expansion.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2022. Amount: $1K, **Funded**.

[11] *Experiences for Teachers on Cyber Security.* **Proposal Developer**. NSF/NSA GenCyber Teacher Summer Camp, 2022. Amount: $140K, **Funded**.

[10] *The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2022. Amount: $1K, **Funded**.

[9] *Crowdsourcing Infrastructure Management System.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2021. Amount: $1K, **Funded**.

[8] *Community Infrastructure Management Application.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2020. Amount: $1K, **Funded**.

[7] *Development of the Pavement Preservation and Rehabilitation Academy.* **Co-PI**. Wirtgen Group - John Deere Company, 2020. Amount: $40K, **Funded**.

[6] *Trade-off Model of Fog-Cloud Computing for Space Network.* **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2020. Amount: $5K, **Funded**.

[5] *Emotion Classification of Users in Social Media.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2019. Amount: $1K, **Funded**.

[4] *Experiences for Teachers on Cyber Security.* **PI**. NSF/NSA GenCyber Teacher Summer Camp, 2019. Amount: $73K, **Funded**.

[3] *Trade-off Model of Fog-Cloud Computing for Space Network.* **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2019. Amount: $5K, **Funded**.

[2] *WV Faculty Mentor Internship Program - Calendar and Special Education Applications.* **Co-PI**. West Virginia Department of Education, 2019. Amount: $85K, **Funded**.

[1] *WV Faculty Mentor Internship Program - Complain Management Application and Security of Applications.* **Co-PI**. West Virginia Department of Education, 2019. Amount: $79K, **Funded**.

Internal Funded Grant Applications

[25] *Augmented Reality as an Aid for Physics Concepts.* **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2022. Amount: $4K, **Funded**.

[24] *Auto Feedback System based on Artificial Intelligence for Cybersecurity Learners.* **PI**. Faculty Summer Research, Marshall University, 2022. Amount: $2K, **Funded**.

[23] *Automatic Feedback System to Teach Cybersecurity Principles.* **Mentor**. Undergraduate Fall Creative Discovery Scholar and Undergraduate Research Scholar Awards, Marshall University, 2022. Amount: $1,750.00, **Funded**.

[22] *Marshall University Presentation Center: Developing A Campus-Wide Resource to Enhance Communication Fluency Across the Curriculum.* **Co-PI**. Hedrick Program Grant for Teaching Innovation, Marshall University, 2022. Amount: $5K, **Funded**.

[21] *The Cybersecurity Packet Control Simulator.* **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2022. Amount: $4K, **Funded**.

[20] *The Cybersecurity Packet Control Simulator: The Effect of Visual Learning Tools on Retention of Information in Computer Science.* **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2022. Amount: $5K, **Funded**.

[19] *The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences.* **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2022. Amount: $5K, **Funded**.

[18] *Crowdsourcing Infrastructure Management System.* **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2021. Amount: $5K, **Funded**.

[17] *Enhancing STEM Education with Augmented Reality.* **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2020. Amount: $4K, **Funded**.

[16] *A Smart Therapy Tool for Feeding and Speech Disorder Detection.* **PI**. John Marshall University Summer Scholars Awards, Marshall University, 2019. Amount: $6.5K, **Funded**.

[15] *Artificial Intelligence and Integrated Fog-Cloud Computing based Matching Algorithm for Carpooling.* **PI**. Faculty Summer Research, Marshall University, 2019. Amount: $2K, **Funded**.

[14] *Augmented Reality based Application for Data Structure.* **Mentor**. Undergraduate Fall Research Scholar Awards, Marshall University, 2019. Amount: $250.00, **Funded**.

[13] *Development of Story-Assisted Platform for Early Childhood for Coding.* **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2019. Amount: $5K, **Funded**.

[12] *Emotion Classification of Users based on the Comments and Emojis in Social Media.* **Mentor**. Undergraduate Fall Creative Discovery Scholar and Undergraduate Research Scholar Awards, Marshall University, 2019. Amount: $1,750.00, **Funded**.

[11] *Profile Analysis on Cryptocurrency Investors and Social Engineering on their Prices.* **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2019. Amount: $4K, **Funded**.

[10] *Quinlan Endowment Faculty Travel*. **PI**. Quinlan Endowment Faculty Travel, Marshall University, 2019. Amount: $500.00, **Funded**.

[9] *Student Research and Innovation Center*. **Co-PI**. Cross-disciplinary Research Facilitation Grant, Marshall University, 2019. Amount: $500.00, **Funded**.

[8] *Automated IP Reputation Analyzer System*. **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2018. Amount: $4K, **Funded**.

[7] *Automated IP Reputation Analyzer System*. **Mentor**. Undergraduate Fall Creative Discovery Scholar and Undergraduate Research Scholar Awards, Marshall University, 2018. Amount: $1,750.00, **Funded**.

[6] *Game Embedded Storybook*. **Mentor**. Undergraduate Spring Research Scholar Awards, Marshall University, 2018. Amount: $175.00, **Funded**.

[5] *Placement of Electric Vehicle Charging Sections and Traffic Management*. **PI**. Faculty Summer Research, Marshall University, 2018. Amount: $2K, **Funded**.

[4] *Quinlan Endowment Faculty Travel*. **PI**. Quinlan Endowment Faculty Travel, Marshall University, 2018. Amount: $400.00, **Funded**.

[3] *Suturing Technique Simulation*. **Mentor**. Undergraduate Spring Research Scholar Awards, Marshall University, 2018. Amount: $175.00, **Funded**.

[2] *Research Initiation and Developemnt*. **PI**. Marshall University, 2017. Amount: $100,000.00, **Funded**.

[1] *Self-dynamic Data Center Management to Optimize Search Speed for Media Type Files*. **PI**. Faculty Summer Research, Marshall University, 2017. Amount: $2,000.00, **Funded**.

PUBLICATIONS

Patents and Intellectual Property

[1] *Oral Therapy Tool, System, and Related Methods*. Patent Appl. Serial No. 62/947,264, Regular, United States. (Submitted: November 2019, Application: December 2019, Updated: October 2022, Allowed January 2023).

Books

[1] **Husnu S. Narman** and Mohammed Atiquzzaman. *Carrier Assignment and Packet Scheduling in LTE-A and Wi-Fi*. Dissertation as a Book. LAP LAMBERT Academic Publishing, May 2016, p. 160. ISBN: 9783659891977. URL: `https://www.amazon.com/Carrier-Assignment-Packet-Scheduling-Wi-Fi/dp/3659891975`.

Journals

[14] Neil Loftus* and **Husnu S. Narman**. "Use of Machine Learning in Interactive Cybersecurity and Network Education". In: *Sensors* 23.6 (2023). ISSN: 1424-8220.

[13] Amrit Pal, Abishi Chowdhury, Satakshi, **Husnu S. Narman**, Arkabandhu Chowdhury, and Manish Kumar. "Random Partition based Adaptive Distributed Kernelized SVM for Big Data". In: *IEEE Access* (2022).

[12] Arnob Paul, Md. Hasanul Islam, Md. Shohrab Hossain, and **Husnu S. Narman**. "A novel zone walking protection for secure DNS Server". In: *International Journal of Interdisciplinary Telecommunications and Networking* (2022).

[11] **Husnu S. Narman**, Haroon Malik, and Govind Yatnalkar^{+}. "An Enhanced Ride Sharing Model Based on Human Characteristics, Machine Learning Recommender System, and User Threshold Time". In: *Springer Journal of Ambient Intelligence and Humanized Computing* (2021). **Invited**.

[10] Jinwei Liu, Haiying Shen, Hongmei Chi, **Husnu S. Narman**, Yongyi Yang, Long Cheng, and Wingyan Chung. "A Low-Cost Multi-Failure Resilient Replication Scheme with Data Correlation for High Data Availability in Cloud Storage". In: *IEEE/ACM Transaction on Networking* (2020).

[9] Abishi Chowdhury, Shital A. Raut, and **Husnu S. Narman**. "DA-DRLS: Drift adaptive deep reinforcement learning based scheduling for IoT resource management". In: *Journal of Network and Computer Applications* 138 (May 2019), pp. 51–65.

[8] Ankur Sarker, Haiying Shen, M. Rahman, M. Chowdhury, K. Dey, F. Li, Y. Wang, and **Husnu S. Narman**. "A Review of Sensing and Communication, Human Factors, and Controller Aspects for Information-Aware Connected and Automated Vehicles". In: *IEEE Transactions on Intelligent Transportation Systems* (March 2019).

[7] Kuo-Chi Fang+, **Husnu S. Narman**, Ibrahim Hussein Mwinyi+, and Wook-Sung Yoo. "PPHA-Popularity Prediction Based High Data Availability for Multimedia Data Center". In: *International Journal of Interdisciplinary Telecommunications and Networking* 11.1 (January 2019), pp. 17–29.

[6] Jinwei Liu, Haiying Shen, **Husnu S. Narman**, Z. Lin, and Z. Li. "Popularity-aware Multi-failure Resilient and Cost-effective Replication for High Data Durability in Cloud Storage". In: *IEEE Transactions on Parallel and Distributed Systems* (October 2018).

[5] Jinwei Liu, Haiying Shen, **Husnu S. Narman**, Wingyan Chung, and Zongfang Lin. "A Survey of Mobile Crowdsensing Techniques: A Critical Component for The Internet of Things". In: *ACM Transactions on Cyber-Physical Systems* 2.3 (June 2018).

[4] Jinwei Liu, Haiying Shen, L. Yu, **Husnu S. Narman**, J. Zhai, J. O. Hallstrom, and Y. He. "Characterizing Data Deliverability of Greedy Routing in Wireless Sensor Networks". In: *IEEE Transactions on Mobile Computing* 17.3 (March 2018), pp. 543–559.

[3] **Husnu S. Narman**, Md.Shohrab Hossain, Mohammed Atiquzzaman, and Haiying Shen. "Scheduling Internet of Things Applications in Cloud Computing". In: *Annals of Telecommunications* (February 2017).

[2] **Husnu S. Narman**, Mohammed Atiquzzaman, Mehdi Rahmani-andebili, and Haiying Shen. "Joint and Selective Component Carrier Assignment in LTE-A". In: *Computer Networks* (September 2016).

[1] **Husnu S. Narman**, Md.Shohrab Hossain, and Mohammed Atiquzzaman. "Management and Analysis of Multi Class Traffic in Single and Multi-band Systems". In: *Wireless Personal Communications* 83 (July 2015).

Conferences

[44] Eric Shoemaker+, Haroon Malik, **Husnu S. Narman**, and Jamil Chaudri. "Explaining the Unseen: Leveraging XAI to Enhance the Trustworthiness of Black-Box Models in Performance Testing". In: *International Conference on Future Networks and Communications (FNC)*. Halifax, Nova Scotia, August 2023.

[43] Shawon Kumar Saha, Rafidul Islam Sarker, Md Shohrab Hossain, and **Husnu S. Narman**. "Android Malware Detection Using Incremental Learning Approach". In: *IEEE International Black Sea Conference on Communications and Networking (IEEE BLACKSEACOM)*. Istanbul, Turkey, July 2023.

[42] Josh Maddy*, Eric M. Dillon*, and **Husnu S. Narman**. "Adapting Cybersecurity Teacher Training Camp to Virtual Learning". In: *IEEE Integrated STEM Education Conference (IEEE ISEC)*. **Best Paper Award**. Laurel, Maryland, March 2023.

[41] Josh Maddy* and **Husnu S. Narman**. "Virtual Reality Museum Application for the Arts". In: *IEEE Integrated STEM Education Conference (IEEE ISEC)*. Laurel, Maryland, March 2023.

[40] Shahriar Hassan, Md. Asif Muztaba1, Shohrab Hossain, and **Husnu S. Narman**. "A Hybrid Encryption Technique based on DNA Cryptography and Steganography". In: *IEEE Information Technology, Electronics and Mobile Communication Conference (IEEE IEMCON)*. Virtual, October 2022.

[39] Eric M. Dillon*, Craig Carpenter II*, John Cook*, Thomas D. Wills*, and **Husnu S. Narman**. "A Machine Learning-Based Automatic Feedback System to Teach Cybersecurity Principles to K-12 and College Students". In: *IEEE Global Humanitarian Technology Conference (GHTC)*. Virtual, September 2022.

[38] Neil Loftus*, Cameron Green*, and **Husnu S. Narman**. "The Cybersecurity Packet Control Simulator: CSPCS". In: *IEEE Global Humanitarian Technology Conference (GHTC)*. Virtual, September 2022.

[37] Kazi Kader, Md. Tareque Tahsin, Md. Shohrab Hossain, and **Husnu S. Narman**. "Ransomware Detection Using Binary Classification". In: *IEEE Cyber Science and Technology Congress (IEEE CyberSciTech)*. Virtual, October 2021.

[36] Md. Mahbubur Rahman, Md. Shohrab Hossain, Mohammad Mahfuzul Islam, and **Husnu S. Narman**. "An Energy Efficient Gravitational Model for Tree Based Routing in Wireless Sensor Networks". In: *IEEE Information Technology, Electronics and Mobile Communication Conference (IEEE IEMCON)*. Virtual, October 2021.

[35] Eric Shoemaker*, Harrison Randolph*, James Bryce, and **Husnu S. Narman**. "Designing Crowdsourcing Software to Inform Municipalities About Infrastructure Condition". In: *IEEE International Conference on Smart Cities: Improving Quality of Life Using ICT, IoT, AI (HONET-ICT)*. Virtual, October 2021.

[34] Tarannum Zaki, Ismat Jahan, Md. Shohrab Hossain, and **Husnu S. Narman**. "An IoT-Based Complete Smart Drainage System for a Smart City". In: *IEEE Information Technology, Electronics and Mobile Communication Conference (IEEE IEMCON)*. Virtual, October 2021.

[33] Neamat Al Sarah, Fahmida Yasmin Rifat, Md. Shohrab Hossain, and **Husnu S. Narman**. "An Efficient Android Malware Prediction Using Ensemble Machine Learning Algorithms". In: *International Conference on Mobile Systems and Pervasive Computing (MobiSPC)*. Leuven, Belgium, August 2021.

[32] Nusrath Tabassum, Farhin Faiza Neha, Md. Shohrab Hossain, and **Husnu S. Narman**. "A Hybrid Machine Learning based Phishing Website Detection Technique through Dimensionality Reduction". In: *IEEE International Black Sea Conference on Communications and Networking (IEEE BLACKSEACOM)*. Virtual, May 2021.

[31] Tanjim Munir Dipon, Md. Shohrab Hossain, and **Husnu S. Narman**. "Detecting Network Intrusion through Anomalous Packet Identification". In: *International Telecommunication Networks and Applications Conference (ITNAC)*. Virtual, November 2020.

[30] Md. Khairul Islam, Prithula Hridi, Md. Shohrab Hossain, and **Husnu S. Narman**. "Network Anomaly Detection Using LightGBM: A Gradient Boosting Classifier". In: *International Telecommunication Networks and Applications Conference (ITNAC)*. Virtual, November 2020.

[29] Jarred Carter*, **Husnu S. Narman**, Ozlem Cosgun, and Jinwei Liu. "Trade-off Model of Fog-Cloud Computing for Space Information Networks". In: *IEEE Cloud Summit*. Virtual, October 2020.

[28] Md. Turab Hossain, Md. Shohrab Hossain, and **Husnu S. Narman**. "Detection of Undesired Events on Real-World SCADA Power System Through Process Monitoring". In: *11th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*. Virtual, October 2020.

[27] **Husnu S. Narman**, Cameron Berry*, Alex Canfield*, Logan Carpenter*, Jeremy Giese*, Neil Loftus$^{\pm}$, and Isabella Schrader$^{\pm}$. "Augmented Reality for Teaching Data Structures in Computer Science". In: *IEEE Global Humanitarian Technology Conference (GHTC)*. Virtual, October 2020.

[26] Shohel Khan, Ajoy Das, Md. Shohrab Hossain, and **Husnu S. Narman**. "Prediction of Dengue Infected Areas Using a Novel Blockchain Based Crowdsourcing Framework". In: *11th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*. Virtual, October 2020.

[25] Kanimozhi Kalaichelavan^{+}, Haroon Malik, **Husnu S. Narman**, and Sreehari Sreenath^{+}. "What People Complain about Drone Apps? A Large-Scale Empirical Study of Google Play Store Reviews". In: *11th International Conference on Ambient Systems, Networks and Technologies (ANT)*. Warsaw, Poland, April 2020.

[24] Renascence Tarafder Prapty, Shuhana Azmin, Md. Shohrab Hossain, and **Husnu S. Narman**. "Preventing Session Hijacking using Encrypted One-Time-Cookies". In: *Wireless Telecommunications Symposium (WTS)*. Virtual, April 2020.

[23] Sreehari Sreenath^{+}, Haroon Malik, **Husnu S. Narman**, and Kanimozhi Kalaichelavan^{+}. "Assesment and Use of Unmanned Aerial Vehicle for Civil Structural Health Monitoring". In: *3rd International Conference on Emerging Data and Industry 4.0 (EDI40)*. Warsaw, Poland, April 2020.

[22] Govind Yatnalkar$^+$, **Husnu S. Narman**, and Haroon Malik. "An Enhanced Ride Sharing Model Based on Human Characteristics and Machine Learning Recommender System". In: *3rd International Conference on Emerging Data and Industry 4.0 (EDI40)*. **Best Paper Award**. Warsaw, Poland, April 2020.

[21] **Husnu S. Narman** and Alymbek Damir Uulu*. "Impacts of Positive and Negative Comments of Social Media Users to Cryptocurrency". In: *IEEE International Conference on Computing, Networking and Communications (ICNC)*. Big Island, HI, February 2020.

[20] Jared Lee Lewis*, Geanina F. Tambaliuc*, **Husnu S. Narman**, and Wook-Sung Yoo. "IP Reputation Analysis of Public Databases and Machine Learning Techniques". In: *IEEE International Conference on Computing, Networking and Communications (ICNC)*. Big Island, HI, February 2020.

[19] Anh Nguyen*, Greg Weed*, and **Husnu S. Narman**. "Oral Therapeutic Tool for Speech and Feeding Therapies". In: *IEEE Global Communications Conference (GLOBECOM)*. Waikoloa, HI, December 2019.

[18] Md. Ishtiaq Ashiq, Protick Bhowmick, Md. Shohrab Hossain, and **Husnu S. Narman**. "Domain Flux-based DGA Botnet Detection Using Feedforward Neural Network". In: *IEEE Military Communication Conference (MILCOM)*. Norfolk, VA, November 2019.

[17] Mainuddin Ahmad Jonas, Risul Islam, Md. Shohrab Hossain, **Husnu S. Narman**, and Mohammed Atiquzzaman. "An Intelligent System for Preventing SSL Stripping-based Session Hijacking Attacks". In: *IEEE Military Communication Conference (MILCOM)*. Norfolk, VA, November 2019.

[16] Govind Yatnalkar$^+$ and **Husnu S. Narman**. "A Matching Model for Vehicle Sharing Based on User Characteristics and Tolerated-Time". In: *IEEE International Conference on Smart Cities: Improving Quality of Life Using ICT, IoT, AI (HONET-ICT)*. Charlotte, NC, October 2019.

[15] **Husnu S. Narman**, Alymbek D. Uulu*, and Jinwei Liu. "Profile Analysis for Cryptocurrency in Social Media". In: *IEEE International Symposium on Signal Processing and Information Technology (ISSPIT)*. Louisville, KY, December 2018, pp. 229–234.

[14] Govind Yatnalkar$^+$ and **Husnu S. Narman**. "Survey on Wireless Charging and Placement of Stations for Electric Vehicles". In: *IEEE International Symposium on Signal Processing and Information Technology (ISSPIT)*. Louisville, KY, December 2018, pp. 526–531.

[13] I. H. Mwinyi$^+$, **Husnu S. Narman**, K. Fang$^+$, and Wook-Sung Yoo. "Predictive self-learning content recommendation system for multimedia contents". In: *Wireless Telecommunications Symposium (WTS)*. Phoenix, AZ, April 2018.

[12] Jinwei Liu, Haiying Shen, and **Husnu S. Narman**. "CCRP: Customized Cooperative Resource Provisioning for High Resource Utilization in Clouds". In: *IEEE International Conference on Big Data*. Washington, DC, August 2016.

[11] **Husnu S. Narman** and Mohammed Atiquzzaman. "Primary Component Carrier Assignment in LTE-A". In: *EAI QSHINE*. Seoul, South Korea, July 2016.

[10] **Husnu S. Narman** and Mohammed Atiquzzaman. "Analysis of Joint and Partial Component Carrier Assignment Techniques in LTE and LTE-A". In: *IEEE Global Communications Conference (GLOBECOM)*. San Diego, CA, December 2015.

[9] **Husnu S. Narman** and Mohammed Atiquzzaman. "Selective Periodic Component Carrier Assignment Technique in LTE and LTE-A Systems". In: *IEEE Global Communications Conference (GLOBECOM)*. San Diego, CA, December 2015.

[8] **Husnu S. Narman** and Mohammed Atiquzzaman. "Energy aware scheduling and queue management for next generation Wi-Fi routers". In: *IEEE Wireless Communications and Networking Conference Workshops (WCNCW)*. New Orleans, LA, March 2015.

[7] **Husnu S. Narman** and Mohammed Atiquzzaman. "Joint and partial carrier components assignment techniques based on user profile in LTE systems". In: *IEEE Wireless Communications and Networking Conference (WCNC)*. New Orleans, LA, March 2015, pp. 983–988.

[6] **Husnu S. Narman** and Mohammed Atiquzzaman. "Carrier Components Assignment Method for LTE and LTE-A Systems Based on User Profile and Application". In: *IEEE GLOBECOM Workshop on Broadband Wireless Access*. Austin, TX, December 2014.

[5] **Husnu S. Narman**, Md. Shohrab Hossain, and Mohammed Atiquzzaman. "DDSS: Dynamic Dedicated Servers Scheduling for multi priority level classes in cloud servers". In: *IEEE International Conference on Communications (ICC)*. Sydney, Australia, June 2014, pp. 3082–3087.

[4] **Husnu S. Narman**, Md. Shohrab Hossain, and Mohammed Atiquzzaman. "h-DDSS: Heterogeneous Dynamic Dedicated Servers Scheduling in cloud computing". In: *IEEE International Conference on Communications (ICC)*. Sydney, Australia, June 2014, pp. 3475–3480.

[3] **Husnu S. Narman**, Md. Shohrab Hossain, and Mohammed Atiquzzaman. "Multi Class Traffic Analysis of Single and multi-band Queuing System". In: *IEEE Global Communications Conference (GLOBECOM)*. Atlanta, GA, December 2013, pp. 1422–1427.

[2] Md. Shohrab Hossain, **Husnu S. Narman**, and Mohammed Atiquzzaman. "A Novel Scheduling and Queue Management Scheme for multi-band Mobile Routers". In: *IEEE International Conference on Communications (ICC)*. Budapest, Hungary, June 2013, pp. 3787–3791.

[1] **Husnu S. Narman**, Turgay Korkmaz, and Suleyman Tek. "Utilizing distance distribution in determining topological characteristics of multi-hop wireless networks". In: *IEEE International Conference on Computing, Networking and Communications (ICNC)*. San Diego, CA, January 2013, pp. 149–154.

Posters

[19] Neil Loftus*, Cameron Green*, Liana Karim$^{\pm}$, Sawyer Slack$^{\pm}$, and **Husnu S. Narman**. "Use of Machine Learning in Networks and Cybersecurity Education". In: *Undergraduate Research Day at the Capitol (URDC)*. Charleston, WV, February 2023.

[18] Josh Maddy* and **Husnu S. Narman**. "Virtual Reality Museum Application for the Arts". In: *Undergraduate Research Day at the Capitol (URDC)*. Charleston, WV, February 2023.

[17] Neil Loftus*, Cameron Green*, and **Husnu S. Narman**. "The Cybersecurity Packet Control Simulator CSPCS". In: *IEEE COMPSAC Annual Student OER Contest*. **Second Best Presentation Award**. June 2022.

[16] Eric M. Dillon*, Craig Carpenter II*, John Cook*, Thomas David Wills*, and **Husnu S. Narman**. "A Machine Learning-Based Automatic Feedback System to Teach Cybersecurity Principles to K-12 and College Students". In: *Marshall Computer Science Symposium on Emerging Technologies and Undergraduate Research Day*. Huntington, WV, April 2022.

[15] Neil Loftus*, Cameron Green*, and **Husnu S. Narman**. "The Cybersecurity Packet Control Simulator CSPCS". In: *Marshall Computer Science Symposium on Emerging Technologies and Undergraduate Research Day*. **Second Best Presentation Award**. Huntington, WV, April 2022.

[14] Josh Maddy*, Eric Dillon*, and **Husnu S. Narman**. "Adapting a GenCyber Teacher Training Camp to Virtual Learning". In: *Marshall Computer Science Symposium on Emerging Technologies and Undergraduate Research Day*. Huntington, WV, April 2022.

[13] Eric Shoemaker* and **Husnu S. Narman**. "Crowdsourcing Infrastructure Management System". In: *Marshall Computer Science Symposium on Emerging Technologies*. **Best Presentation Award**. Huntington, WV, April 2021.

[12] William Coleman* and **Husnu S. Narman**. "Learning Object-Oriented Programming Concepts with Augmented Reality". In: *Undergraduate Research Day at the Capitol (URDC)*. Virtual, March 2021.

[11] William Coleman* and **Husnu S. Narman**. "Enhancing Computer Science Education with Augmented Reality". In: *West Virginia University Summer Undergraduate Research Symposium*. Virtual, July 2020. URL: `https://voicethread.com/myvoice/thread/14889356/92900841`.

[10] Jarred Carter* and **Husnu S. Narman**. "Trade-off Model of Fog-Cloud Computing for Space Information Networks". In: *Undergraduate Research Day at the Capitol (URDC)*. Charleston, WV, February 2020.

[9] James Farley* and **Husnu S. Narman**. "Emotion Classification of Users Based on Posts and Comments in Social Media". In: *Undergraduate Research Day at the Capitol (URDC)*. Charleston, WV, February 2020.

[8] Cameron Berry*, Alex Canfield*, Logan Carpenter*, Jeremy Giese*, and **Husnu S. Narman**. "Augmented Reality in Computer Science Education". In: *Coding and Cybersecurity Summit*. Charleston, WV, December 2019.

[7] Anh Nguyen*, Greg Weed*, and **Husnu S. Narman**. "Oral Therapeutic Tool". In: *Marshall Computer Science Symposium on Emerging Technologies*. Huntington, WV, April 2019.

[6] Alymbek Damir Uulu* and **Husnu S. Narman**. "Education Analysis of Cryptocurrency for Protection from Social Engineering". In: *Marshall Computer Science Symposium on Emerging Technologies*. Huntington, WV, April 2019.

[5] Govind Yatnalkar$^+$ and **Husnu S. Narman**. "Machine Learning Vehicle Sharing & Matching Model Based on User Profiling". In: *Marshall Computer Science Symposium on Emerging Technologies*. Huntington, WV, April 2019.

[4] Geanina Tambaliu*, Jared Lee Lewis*, **Husnu S. Narman**, and Wook-Sung Yoo. "Automated IP Reputation Analyzer". In: *Undergraduate Research Day at the Capitol (URDC) and Cybersecurity and Privacy Day (Sponsored by WV Office of Technology)*. Charleston, WV, February 2019. URL: `https://hsnarman.github.io/POSTER/IPReputation.jpg`.

[3] Alymbek Damir Uulu* and **Husnu S. Narman**. "Education Analysis of Cryptocurrency for Protection from Social Engineering". In: *Cybersecurity and Privacy Day (Sponsored by WV Office of Technology)*. Charleston, WV, February 2019. URL: `https://hsnarman.github.io/POSTER/SocialEngineering.jpg`.

[2] Alex Kacinari*, Chris Murphy*, Derek M Staley*, and **Husnu S. Narman**. "Suturing Wound Simulation". In: *Marshall Computer Science Symposium on Emerging Technologies*. **3rd Best Presentation Award**. Huntington, WV, April 2018.

[1] Charlie Murphy*, Michael B Branard*, Steven D. Gunnels*, and **Husnu S. Narman**. "Game-Story Book". In: *Marshall Computer Science Symposium on Emerging Technologies*. Huntington, WV, April 2018.

Technical Reports

[6] Ibrahim Hussein Mwinyi$^+$, **Husnu S. Narman**, Kuo-Chi Fang$^+$, and Wook-Sung Yoo. *Recommendation System based on Predictive Approach*. Marshall University, October 2017. URL: `https://hsnarman.github.io/TR/17-Predictive-TR-MU-CITE-17-100.pdf`.

[5] **Husnu S. Narman** and Mohammed Atiquzzaman. *User Profile Carrier Components Assignment Method in LTE Systems*. University of Oklahoma, October 2014. URL: `https://hsnarman.github.io/TR/14-User-Profile-TR-OU-TNRL-14-102.pdf`.

[4] **Husnu S. Narman** and Mohammed Atiquzzaman. *Analysis of Static Partial Carrier Components Assignment in LTE Systems*. University of Oklahoma, September 2014. URL: `https://hsnarman.github.io/TR/14-Lte-Cc-Static-TR-OU-TNRL-14-101.pdf`.

[3] **Husnu S. Narman**, Md Shohrab Hossain, and Mohammed Atiquzzaman. *DDSS:Dynamic Dedicated Servers Scheduling for multi priority level classes in cloud servers*. University of Oklahoma, September 2013. URL: `https://hsnarman.github.io/TR/13-cloud-priority-TR-OU-TNRL-13-102.pdf`.

[2] Md Shohrab Hossain, **Husnu S. Narman**, and Mohammed Atiquzzaman. *Scheduling and Queue Management of Multi-band Mobile Routers*. University of Oklahoma, February 2013. URL: `https://hsnarman.github.io/TR/13-Multi_band_multi_TR-OU-TNRL-13-101.pdf`.

[1] **Husnu S. Narman**, Md Shohrab Hossain, and Mohammed Atiquzzaman. *Multi-Class Traffic Analysis of Single and Multi-band Queuing System*. University of Oklahoma, October 2012. URL: `https://hsnarman.github.io/TR/12-Multi_Band_Single_TR-OU-TNRL-12-101.pdf`.

Press and News

[2] "Summer Camp on Robotics and Cybersecurity". In: *WCHS TV*. Huntington, WV, June 2019. URL: https://hsnarman.github.io/media/news/wsaz/camp2019.mp4.

[1] "Summer Camp on Robotics". In: *WCHS TV*. Huntington, WV, June 2018. URL: https://hsnarman.github.io/media/news/wsaz/camp2018.mp4.